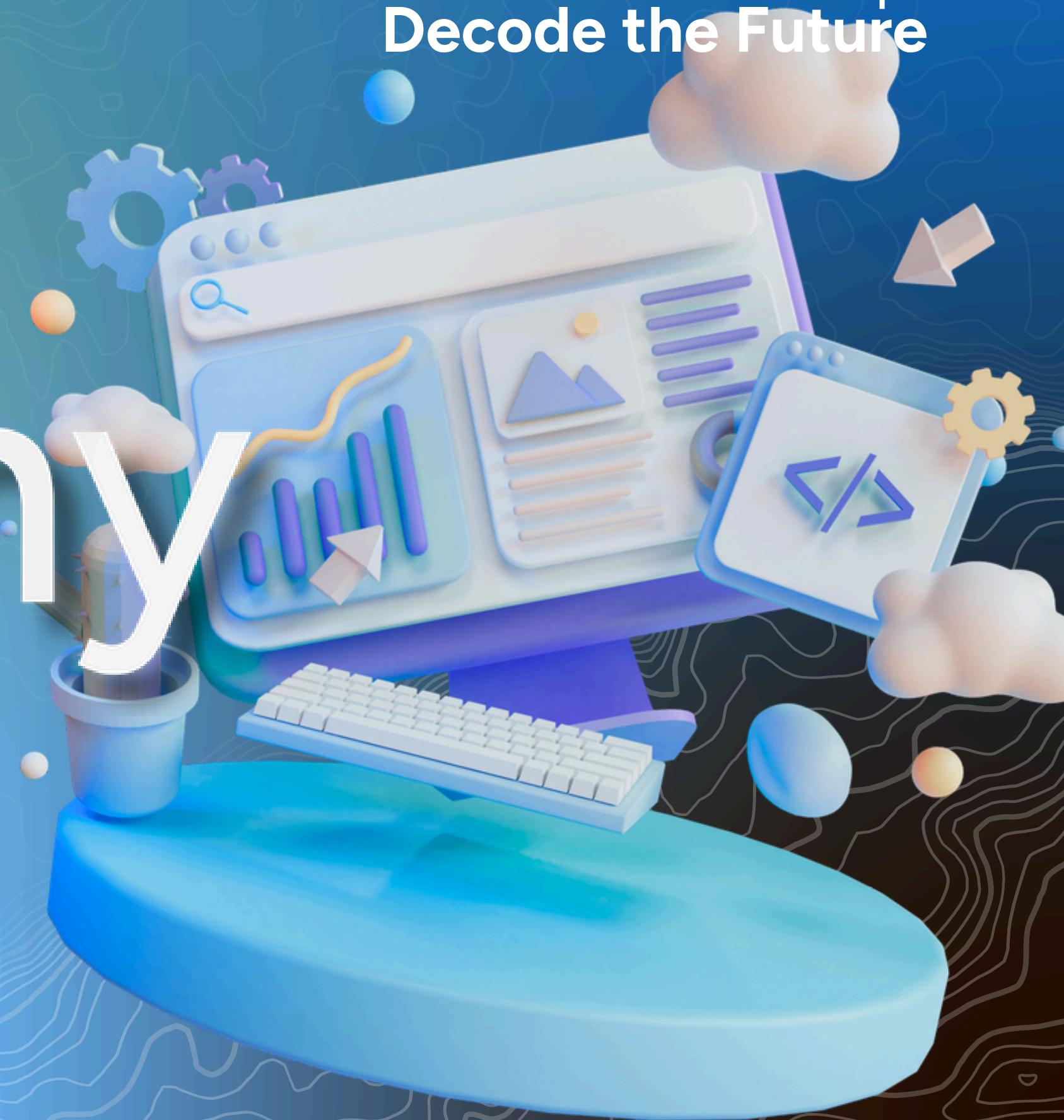


**2026**

Code the Map  
Decode the Future

# MAPID Academy

**WebGIS Development Bootcamp**





# Biodata Diri

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**Dzikri N**  
Data Lead/Analyst



## Teknik Geodesi dan Geomatika

Institut Teknologi Bandung



## Data Lead

PT. Multi Areal Planing Indonesia



## GIS Related Project

- Route Optimization for Logistics
- Managing assets intelligently through spatial monitoring and predictive insights.
- Spatial Business Expansion & Performance Analysis



<https://www.linkedin.com/in/dzikri-nashrul-j/>

# Outline

## GIS Fundamental

### Materials:

1. GIS Definition & 3S Concept
2. Model & GIS Structure
3. GIS Data Source & ToU
4. GIS as A Technology
5. Spatial Analytics (Basic Knowledge)
6. Introduction to GEO MAPID

### Hands-On:

1. Data Extraction from Certain Sources
2. Basic Open Software (Desktop Version)
3. Introduction to GEO MAPID

# MAPID Academy

## WebGIS Development Bootcamp

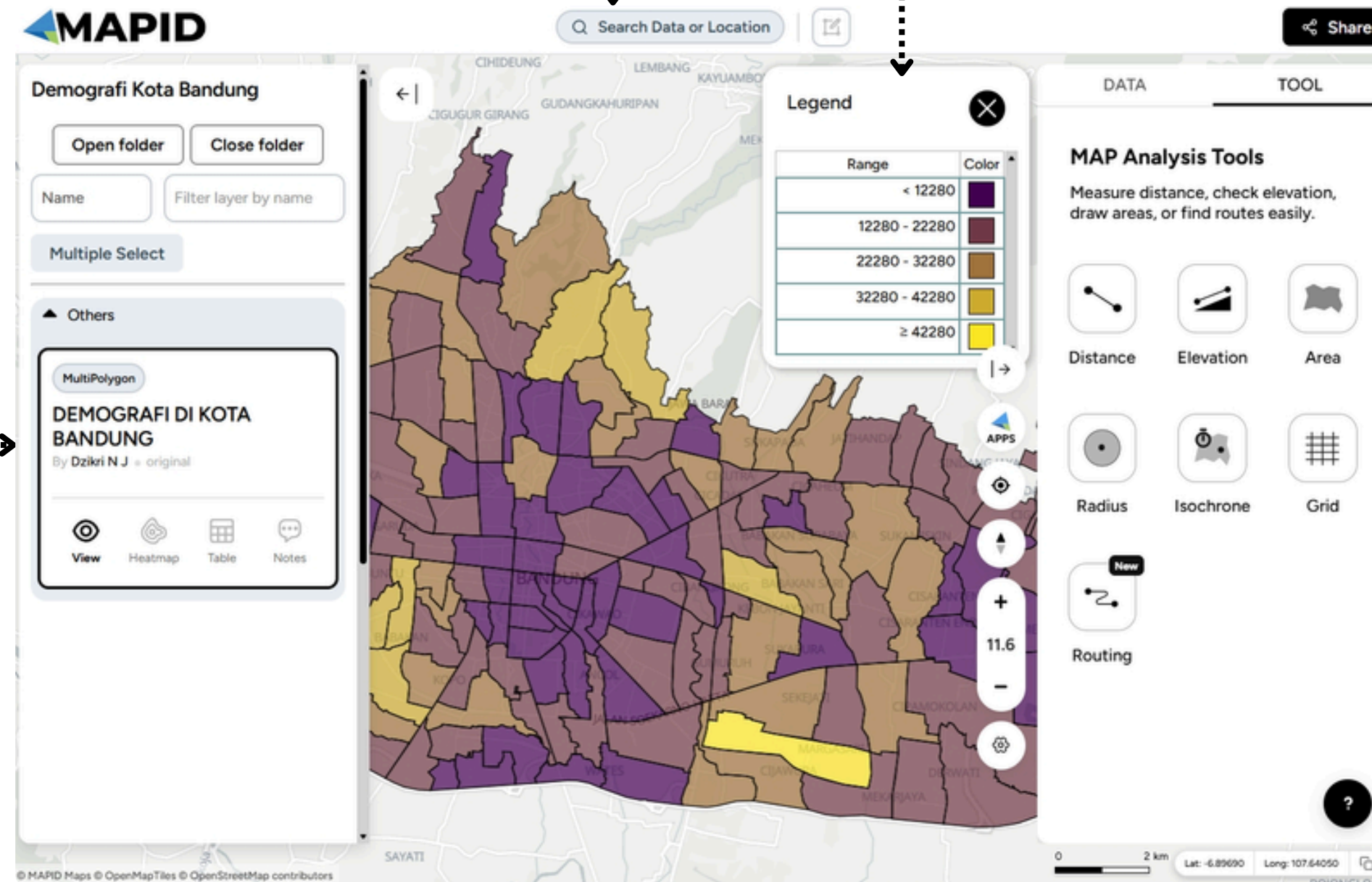
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03



*You can expand location search capabilities in your WebGIS*

*Certain information should be specified in more detail, such as color gradient changes that represent data levels*



*Here you can present the layers that will be displayed in the WebGIS*

*If you imagine your WebGIS could encompass various analyses, you can add several spatial analysis features.*

# Let's start from the end!

*Sometimes in certain WebGIS, scale information and coordinate type are important to be conveyed*





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# Market Size

*-point of view*

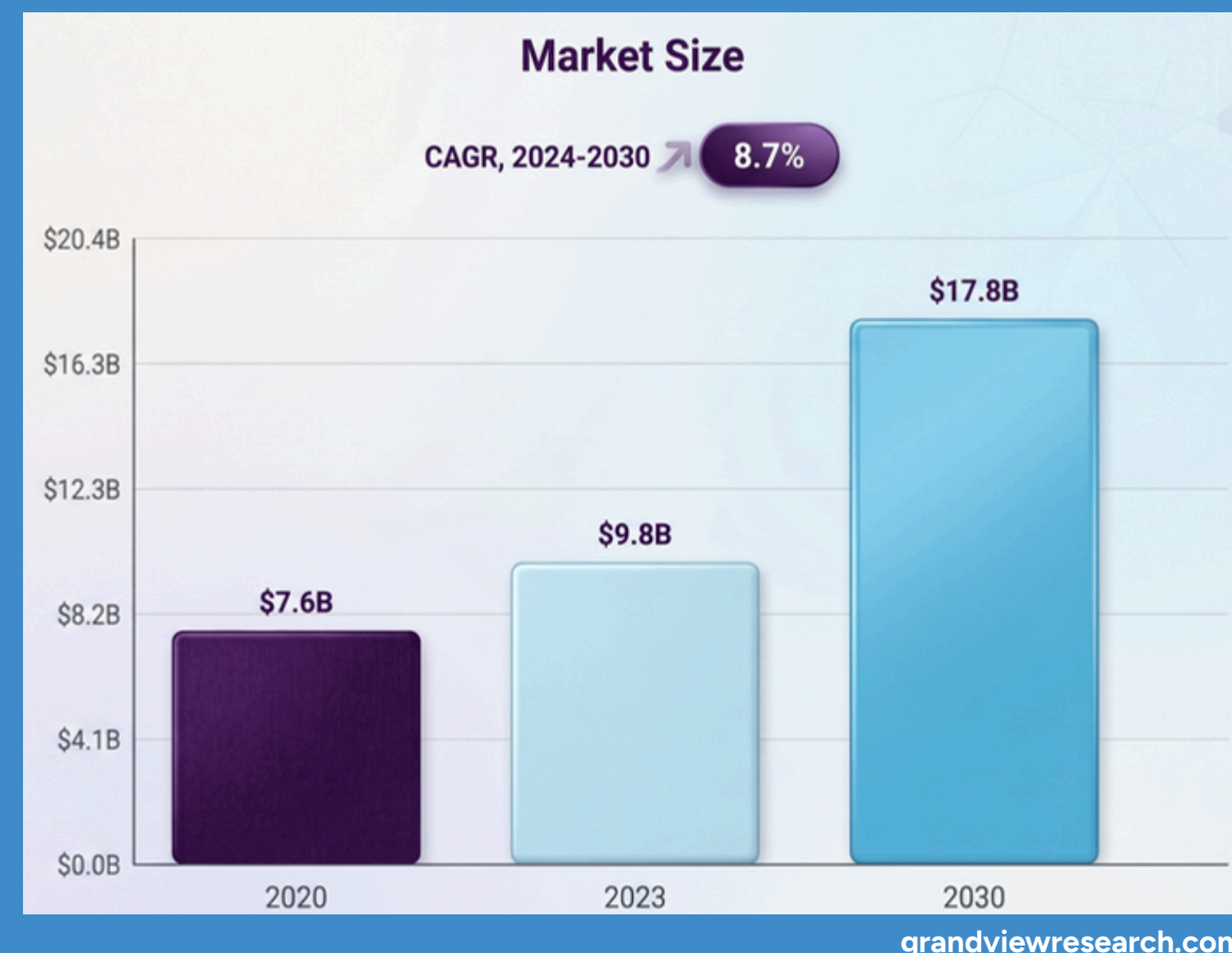
The Geographic Information System Market *is projected to reach USD 17.76 Billion by 2030* from USD 9.80 billion in 2023 at a CAGR of 8.7% during the forecast period.

**Driver:** Development smart cities and urbanization

**Restraint:** Geospatial data barriers and high cost

**Opportunity:** Development of 4D GIS Software and AR

**Challenge:** Integration with other technologies (AI, big data, etc)



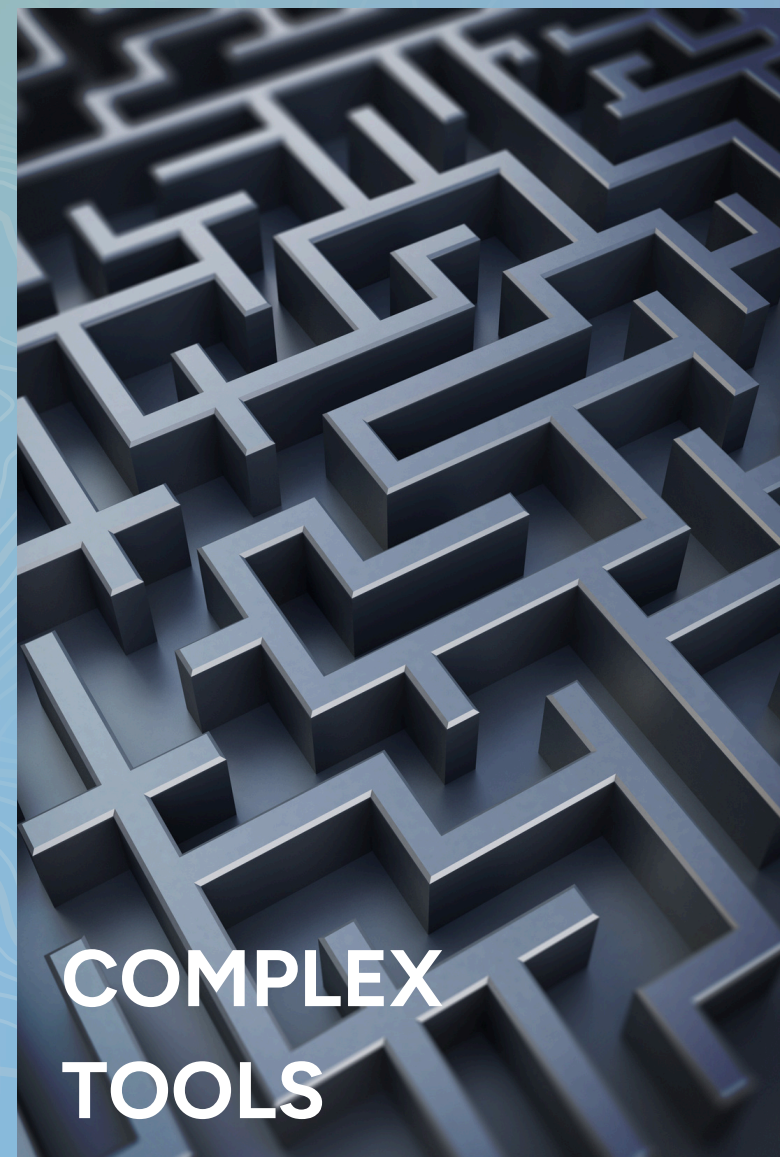
*WebGIS experts who blend tech and spatial insight are in high demand.*



# What's GIS in Digital Age?

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Let's see the problems first!





# GIS Timeline

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## Kelahiran GIS

1963: GIS Pertama Dunia! Roger Tomlinson di Kanada bikin CGIS, sistem komputer pertama yang bisa mengumpulkan, menyimpan, dan analisis data lahan. Ini titik awal GIS yang kita kenal.

1960-an - 1980-an

## Akar Konseptual

Sebelum 1960

Peta Pertama Manusia: Jauh sebelum komputer, manusia sudah bikin peta (di gua, tanah liat) untuk tahu posisi dan wilayah. Ini ide awal tentang lokasi.

1990-an - 2000-an

## GIS Makin Meluas & Era Open Source

- GIS di Internet: Peta mulai bisa diakses lewat browser web. Ini awal mula Web GIS.
- 2002: Lahirnya QGIS! Ini dia! Gary Sherman memulai proyek Quantum GIS (QGIS) sebagai open source GIS desktop.
- 2004: Google Earth & OpenStreetMap (OSM)

2010-an - Sekarang  
**Era Modern**

GIS kini ada di setiap ponsel kita, dan terus berkembang menjadi "otak" cerdas yang mengintegrasikan data real-time dari IoT dan wawasan AI untuk memahat keputusan strategis.



# See The Definitions

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*“A computer-based system to aid in the collection, maintenance, storage, analysis, output, and distribution of spatial data and information.”*

*“A broad range of computer-based tools for capturing, storing, generating, analyzing, and displaying geographic data in support of integrated decision making.”*

Geographic

*Spatial association of data with geographic location*

Informatic

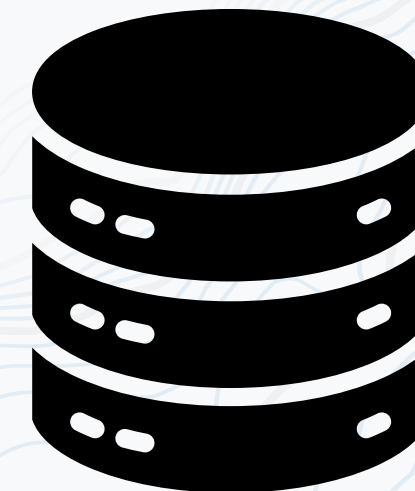
*Structured and well-organized data*

System

*Integrated set of interconnected components*

WebGIS

=



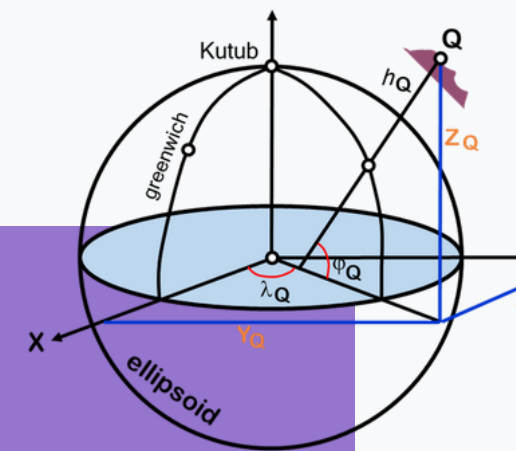
(Geographic Information System)

Spatial Database



# Spatial Data or GIS Data?

Code the Map  
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### Spatial

|              |              |             |
|--------------|--------------|-------------|
| TALIKA<br>12 | TALIKA<br>14 | DRANA<br>16 |
| OXFORD COURT |              |             |
| 11           | 13           | A ST.<br>15 |

MAP

### Spatial

Address

### Nonspatial

Name

Value

### Spatial

Area

|                |        |      |     |
|----------------|--------|------|-----|
| 12 OXFORD CRT. | TALIKA | 5000 | 600 |
| 14 OXFORD CRT. | TALIKA | 7000 | 600 |
| 16 OXFORD CRT. | DRANA  | 6000 | 700 |

DATABASE

Geometry

Coordinate  
Reference

Attribute

Spatial  
relationships

## Spatial Data

Data that describes the **location of the data within a given 'space'**

Geospasial Data: *The spatial aspects of an object or event encompass its place, location, and position relative to, below, on, or above the Earth's surface, expressed in a specific coordinate system.*

Is GIS the same as a map?

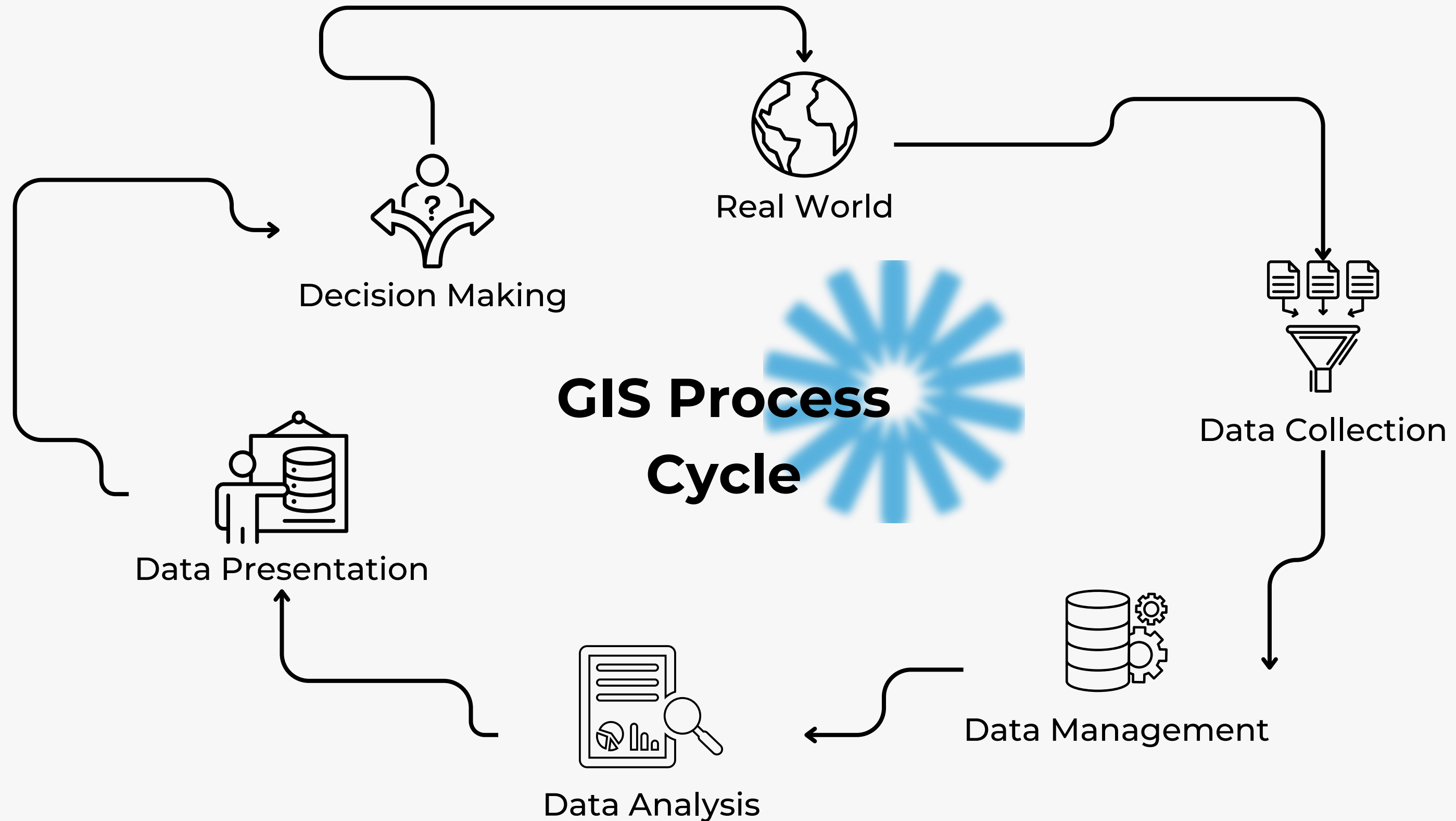
Does GIS only contain spatial data?

Does GIS require a database?



# Siklus Proses SIG

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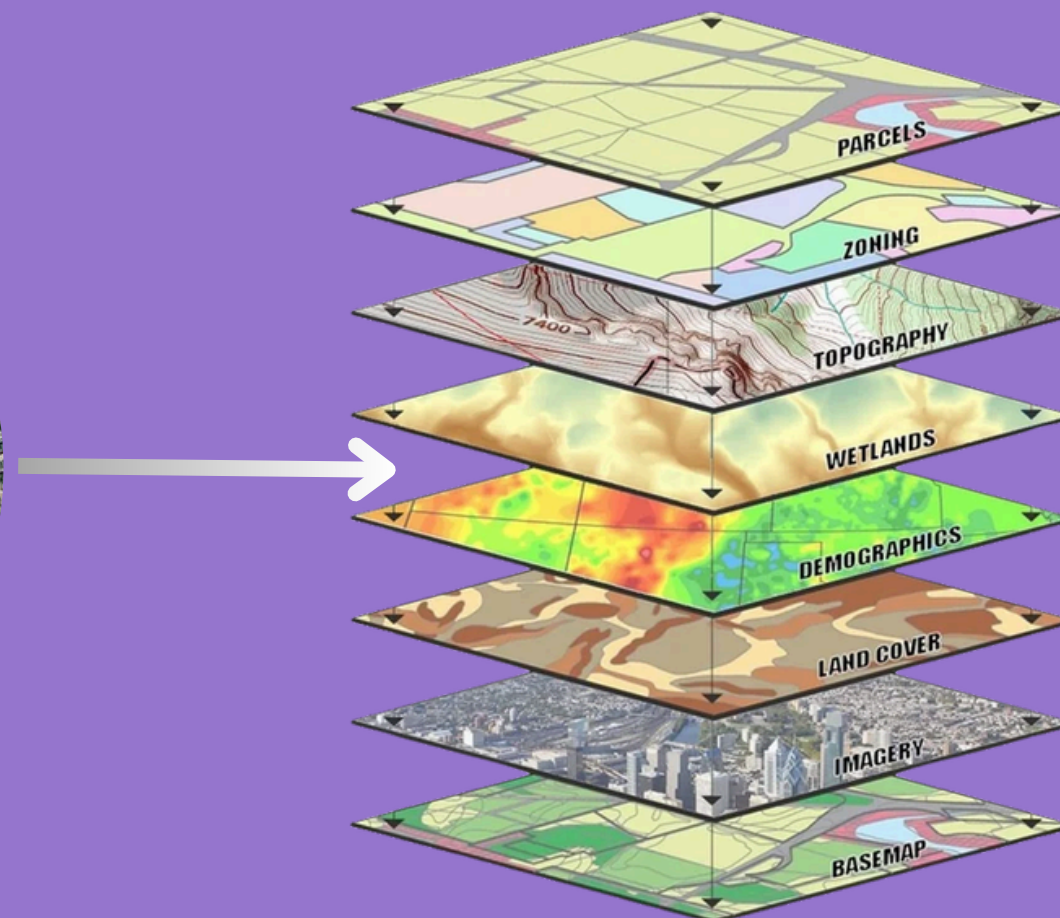


# "3S" dalam SIG

*Real World*



*Layer*



- **System** (Teknologi): Ini alatnya (software, hardware, data) yang bikin kita bisa mengelola informasi lokasi.
- **Science** (Ilmu): Ini konsep dan teorinya (misalnya, akurasi peta) yang memastikan alat itu bekerja benar dan bisa diandalkan.
- **Studies** (Aplikasi): Ini penerapannya untuk bantu masalah manusia dan memahami dampaknya (misalnya, penataan kota, disaster, dll).

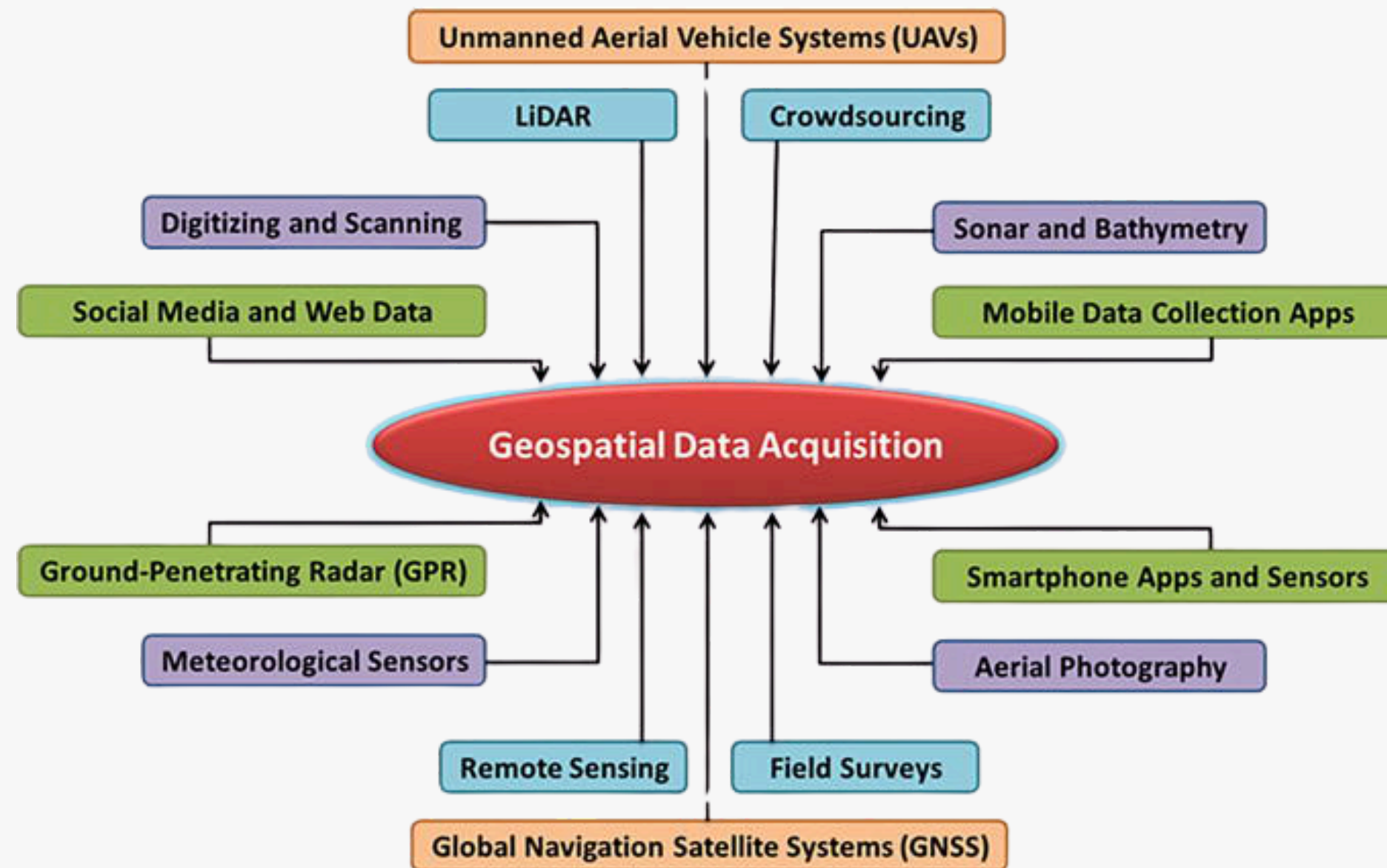
QnA:

- Sebagai sebuah "Sistem", apa kaitannya antara layer dan basis data dalam SIG?
- Bagaimana merepresentasikan *real world* ke dalam SIG?



# Model Data

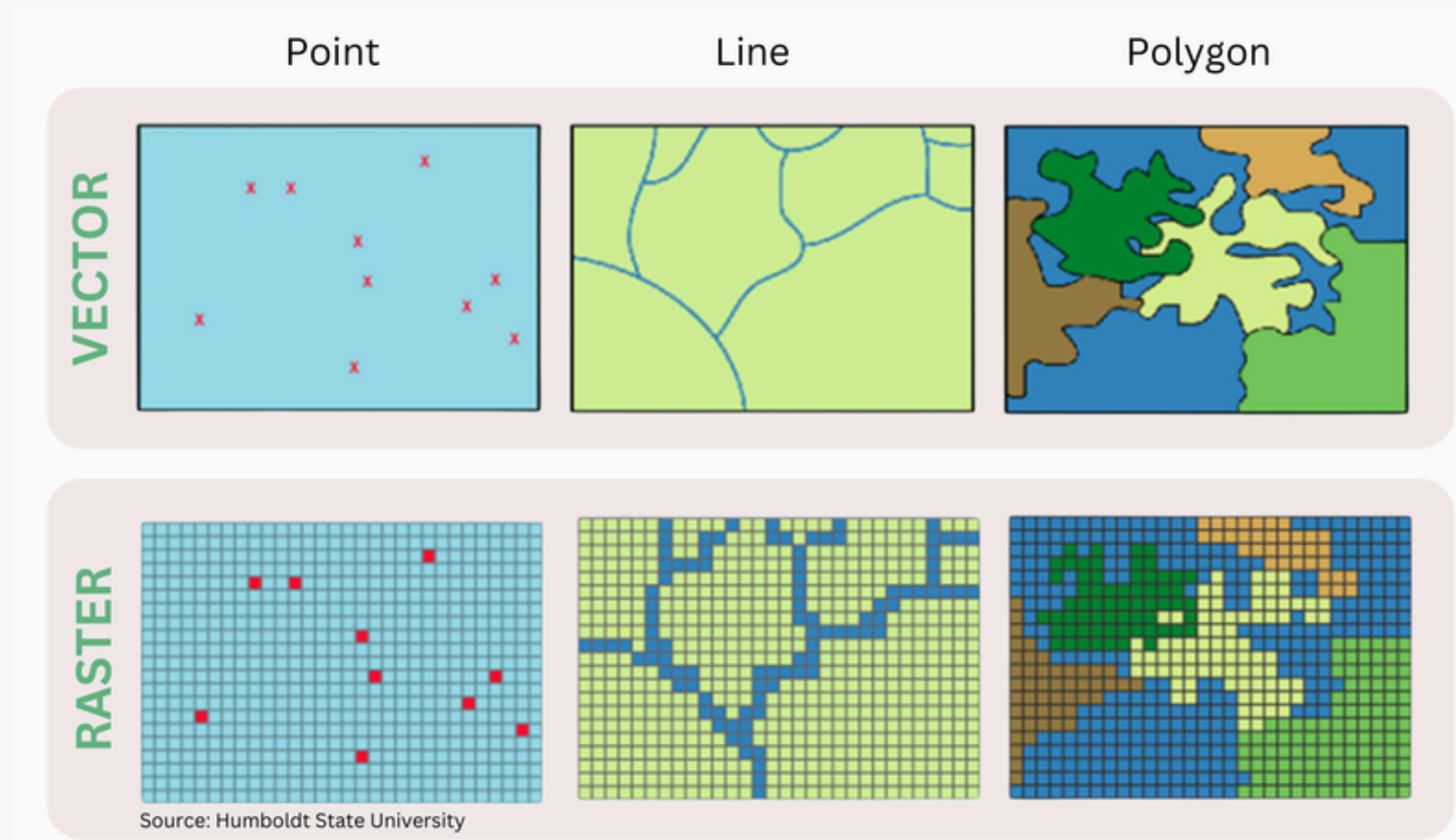
Code the Map  
Decode the Future



- Semua informasi geografis diubah dan disimpan dalam format digital agar bisa diproses komputer.
- Proses Bervariasi: Cara digitalisasinya tergantung alat, seperti GPS untuk titik, satelit untuk citra, atau scanning peta lama.
- Terdapat dua model representasi data spasial, yaitu **Vektor** dan **Raster**

Sumber: <https://www.intechopen.com/>

# Vector vs Raster



## Vector

- Berbasiskan data titik koordinat (x,y).
- Setiap obyek direpresentasikan dalam bentuk unsur geografis:
  - **point or Node** : 0-dimension, single x,y coordinate pair, zero area
  - **line (arc) segment**: 1-dimension, two (or more) connected x,y coordinates
  - **polygon**: 2-dimensions, three or more ordered and connected x,y coordinates , first and last x,y pairs are the same encloses an area
- Disimpan dalam struktur **Topologi & Spaghetti**

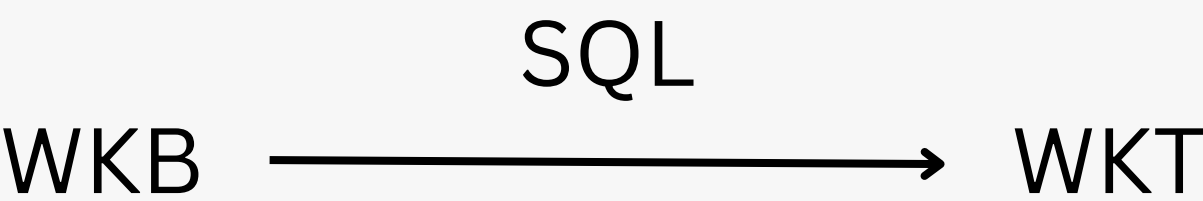
## Raster

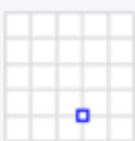
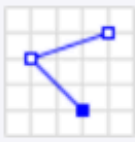
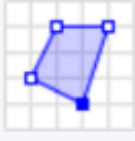
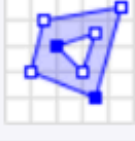
- Setiap obyek direpresentasikan dalam bentuk **pixel** (picture element).
- Pixel tersebut disusun dalam bentuk **baris** dan **kolom** (grid)
- Setiap pixel memiliki nilai data yang merepresentasikan obyek/fenomena geografisnya. Nilai ini umumnya merepresentasikan warna dari obyek/fenomena yang direkamnya)



# About Vector Data

"Features as Well Known Text (WKT) Format"  
Spatial data is usually stored as "binary" data in a database and in files



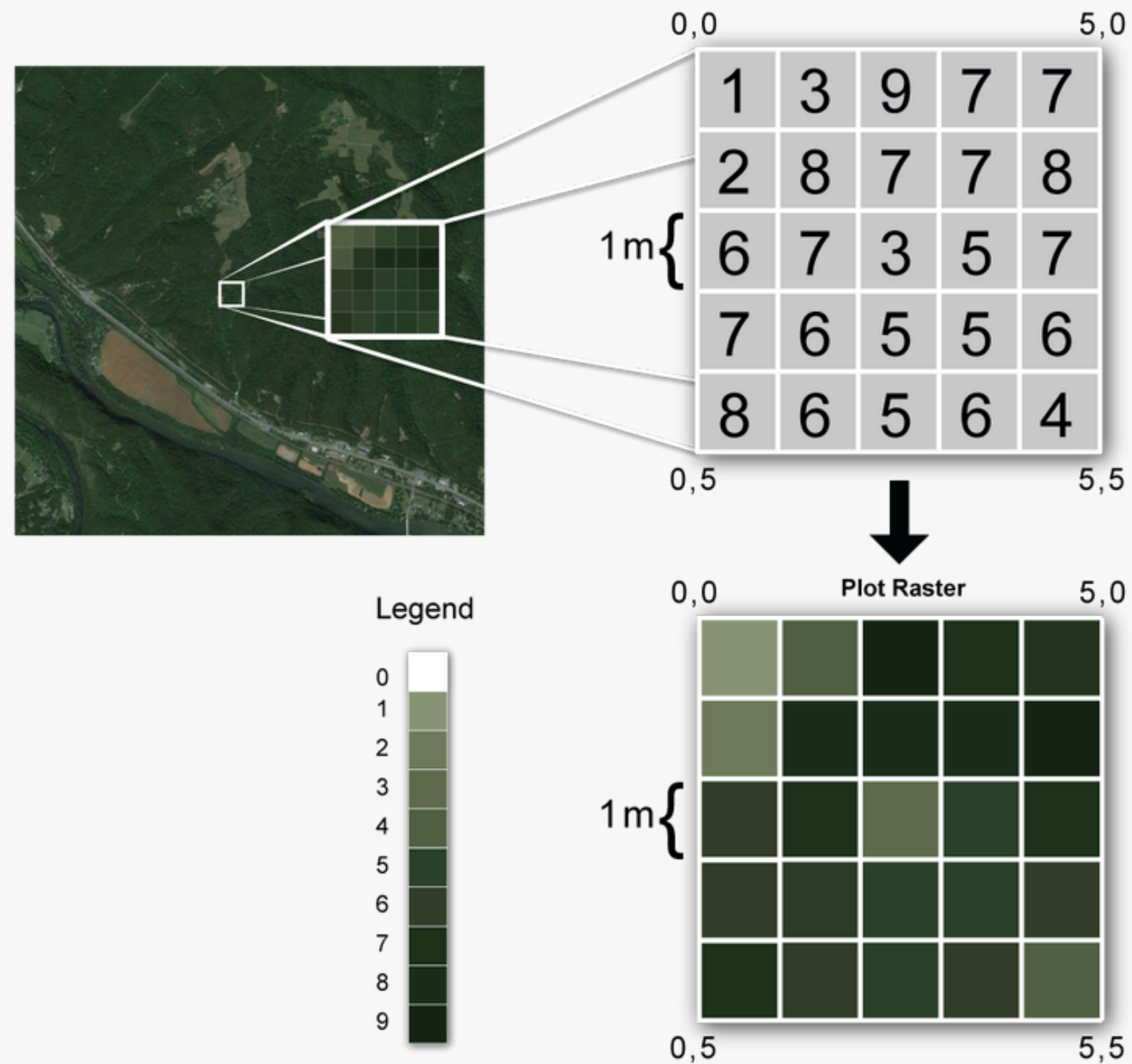
| Type       | Examples                                                                              |                                                                                          |
|------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Point      |  | <code>POINT (30 10)</code>                                                               |
| LineString |  | <code>LINESTRING (30 10, 10 30, 40 40)</code>                                            |
| Polygon    |  | <code>POLYGON ((30 10, 40 40, 20 40, 10 20, 30 10))</code>                               |
|            |  | <code>POLYGON ((35 10, 45 45, 15 40, 10 20, 35 10), (20 30, 35 35, 30 20, 20 30))</code> |

Cari tahu terkait struktur arc-node (topologi) dan spaghetti!

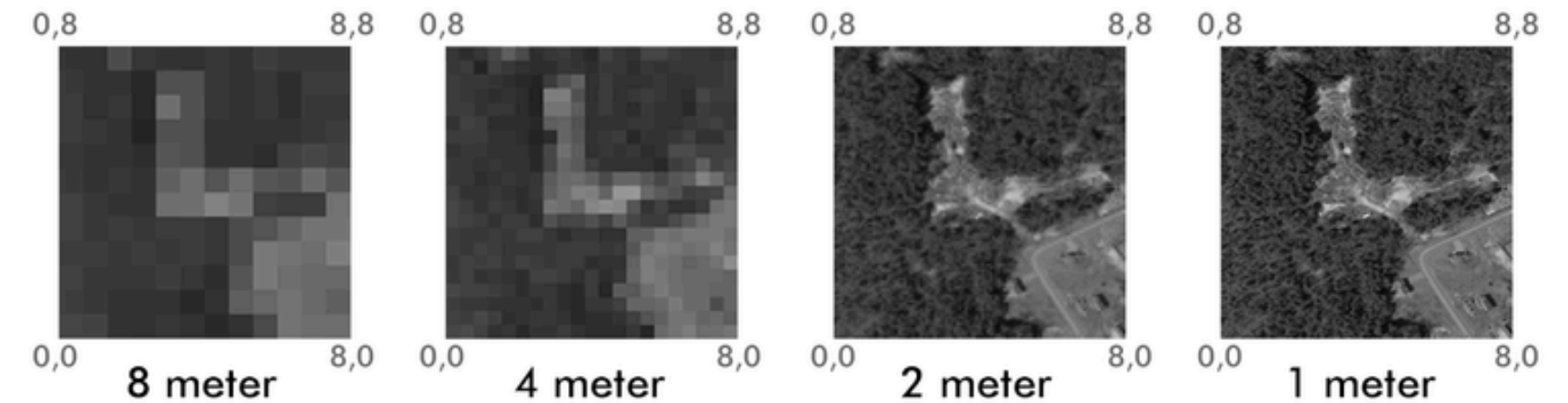
# About Raster Data

Code the Map  
Decode the Future

Sumber: National Ecological Observatory Network (NEON)



Cari tahu perbedaan skala dan resolusi!



Beberapa poin yang perlu diperhatikan,

- Data raster bergantung pada tingkat kerapatan atau jumlah pixel untuk satu area (Resolusi Spasial).
- Resolusi tinggi berarti ukuran pixel semakin kecil dan jumlahnya semakin besar.
- Contoh data: curah hujan dan nilai ketinggian suatu daerah
- Atribut raster mencakup extent, CRS, resolution, dll
- Data raster juga bisa berupa multi/hiperspektral, yaitu citra dengan banyak band.



# GIS Data ETL



## Extract

- Apa yang diekstrak?  
Vektor/Raster/Atribut
- Sumbernya dari mana?  
BIG, OSM, dll?
- Bagaimana caranya?  
unduh/akses API, konversi  
dari berbagai format

## Transform

- Cek sistem koordinat
- Lakukan preprocessing  
data (duplikat, missing  
value, dll)
- Perlu agregasi, merge, dll?
- Lakukan standarisasi  
atribut

## Load

- Muat dalam satu database  
geospasial (cloud atau  
desktop platform?)
- Data yang sudah  
diekstraksi dan ditransform  
dapat dilakukan proses  
**Manipulasi Data**

# Term of Use Data



## LISENCE

### **Kenali Jenis Lisensi:**

- *Terbuka (Open License): Biasanya bebas dipakai, dimodifikasi, dan disebar, ASALKAN sebutkan sumbernya (atribusi). Contoh: OSM, Open Data, dll.*
- *Propietari (Berbayar/Khusus): Ada batasan ketat dalam penggunaan dan distribusi.*



## PRIVACY

### **Jaga Privasi Data:**

- *Data lokasi individu bisa sangat sensitif. Pastikan kamu melindungi privasi mereka.*
- *Patuhi Regulasi. Contohnya:*
  - *GDPR (General Data Protection Regulation)*
  - *UU Perlindungan Data Pribadi (UU PDP).*
  - *Undang-Undang Informasi Geospasial (UU IG) No. 4 Tahun 2011:*



## CREDIBILITY

### **Penting untuk:**

- *Menjaga kredibilitas profesionalmu, gunakan data yang etis tunjukkan profesionalitasmu.*
- *Hindari Masalah Hukum*



# GIS as A Technology

Code the Map  
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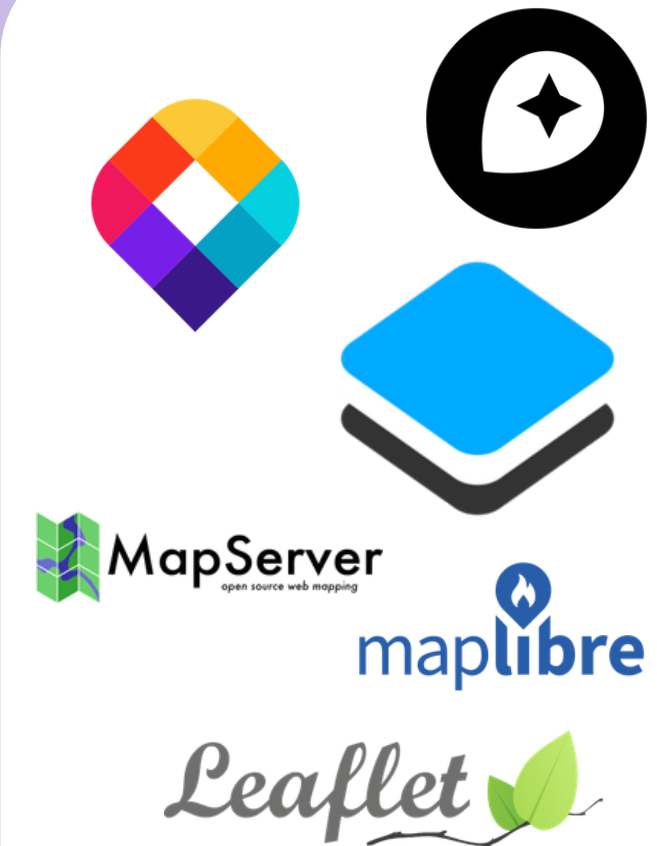
**Desktop GIS  
Platform**



**Cloud (WebGIS)  
Platform**



**Cloud for  
GIS**



**Framework Map  
WebGIS**

# If you catch the open-source

Code the Map  
Decode the Future

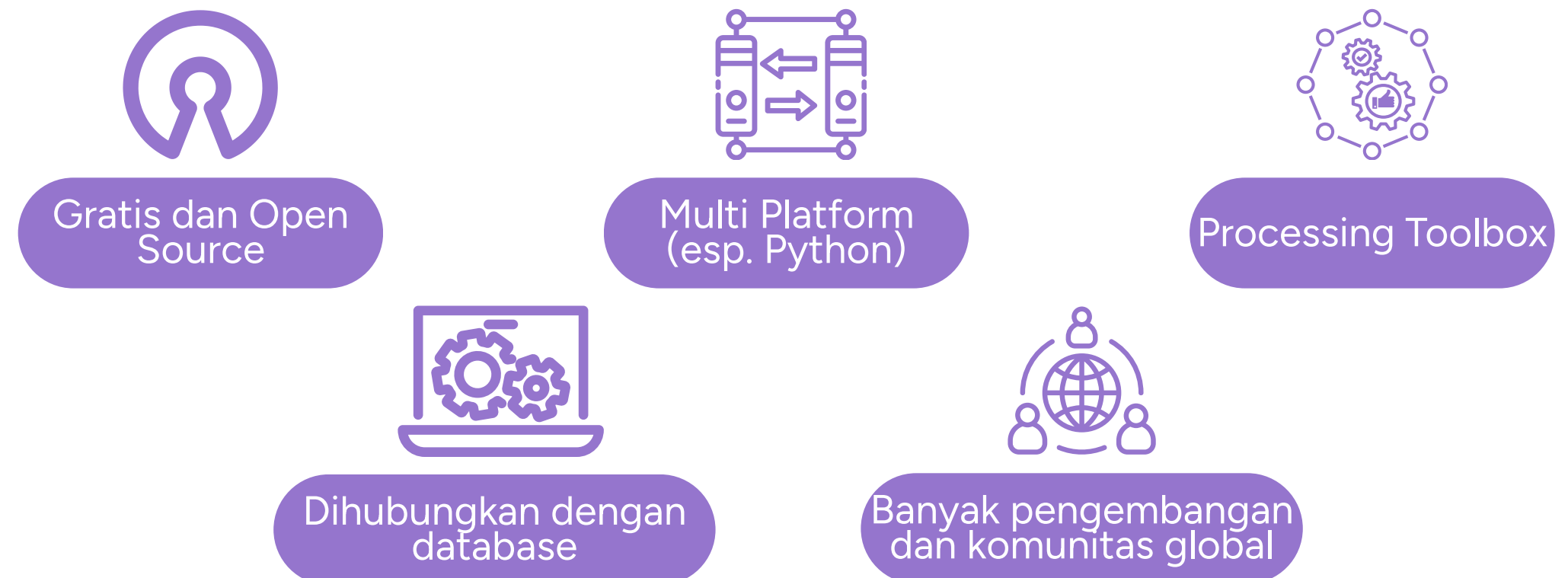


Salah satu jenis perangkat lunak (software) sistem informasi geografis (SIG) berbasis *open-source*. QGIS ini dapat digunakan untuk pengelolaan, pengolahan, dan visualisasi data spasial untuk memberi pengetahuan terkait persoalan-persoalan yang terkait dengan ruang.

## Supported Data di QGIS

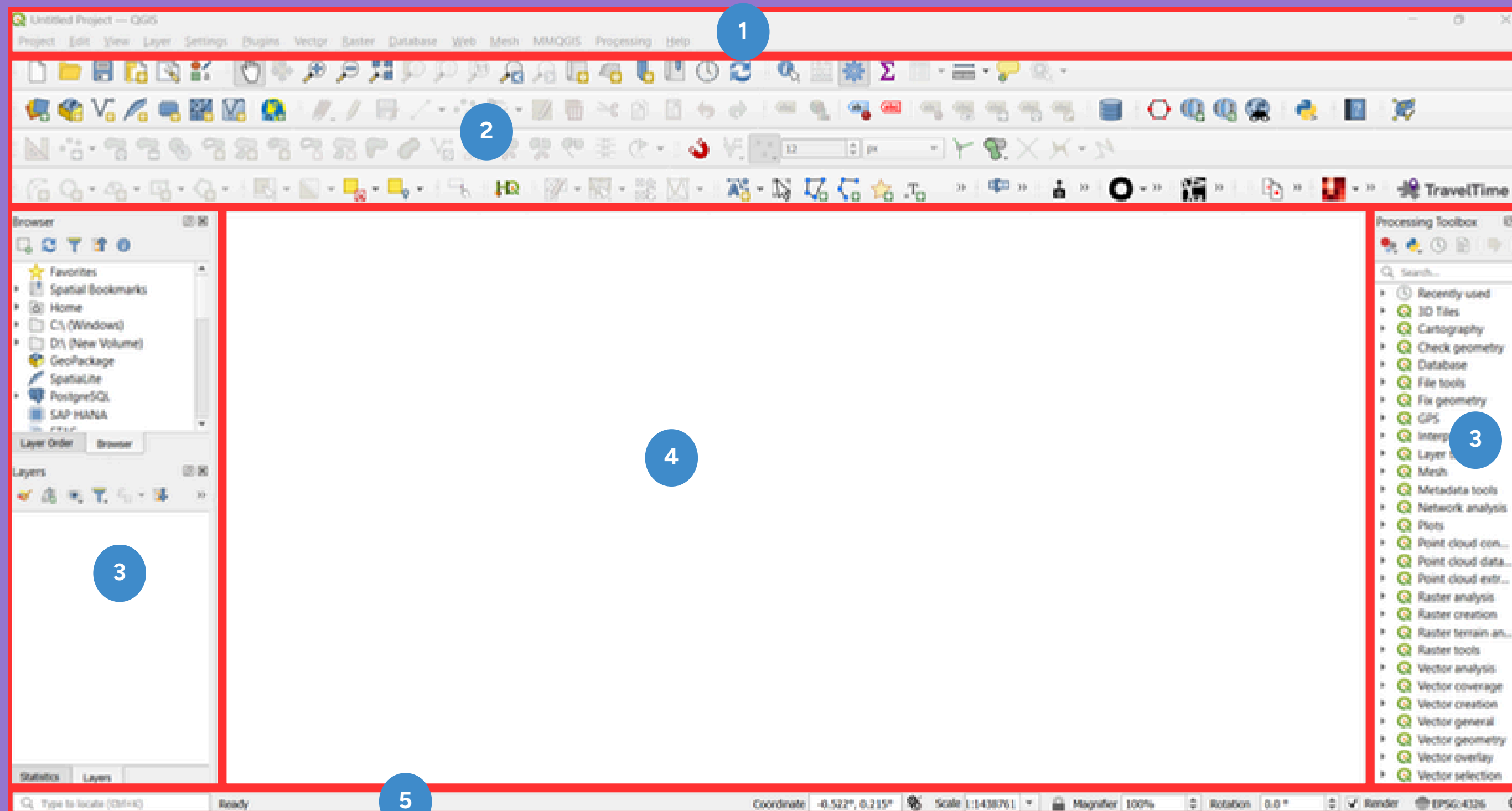
- Vector Data (GeoJSON, ESRI Shapefile, OpenStreetMap, etc)
- Raster Data (GeoTIFF, ASCII, etc)
- Delimited Text
- GPS
- SpatiaLite
- PostgreSQL
- WMS/WMTS
- etc

## Why QGIS?





# QGIS Interface



1.Menu Bar

2.Toolbar

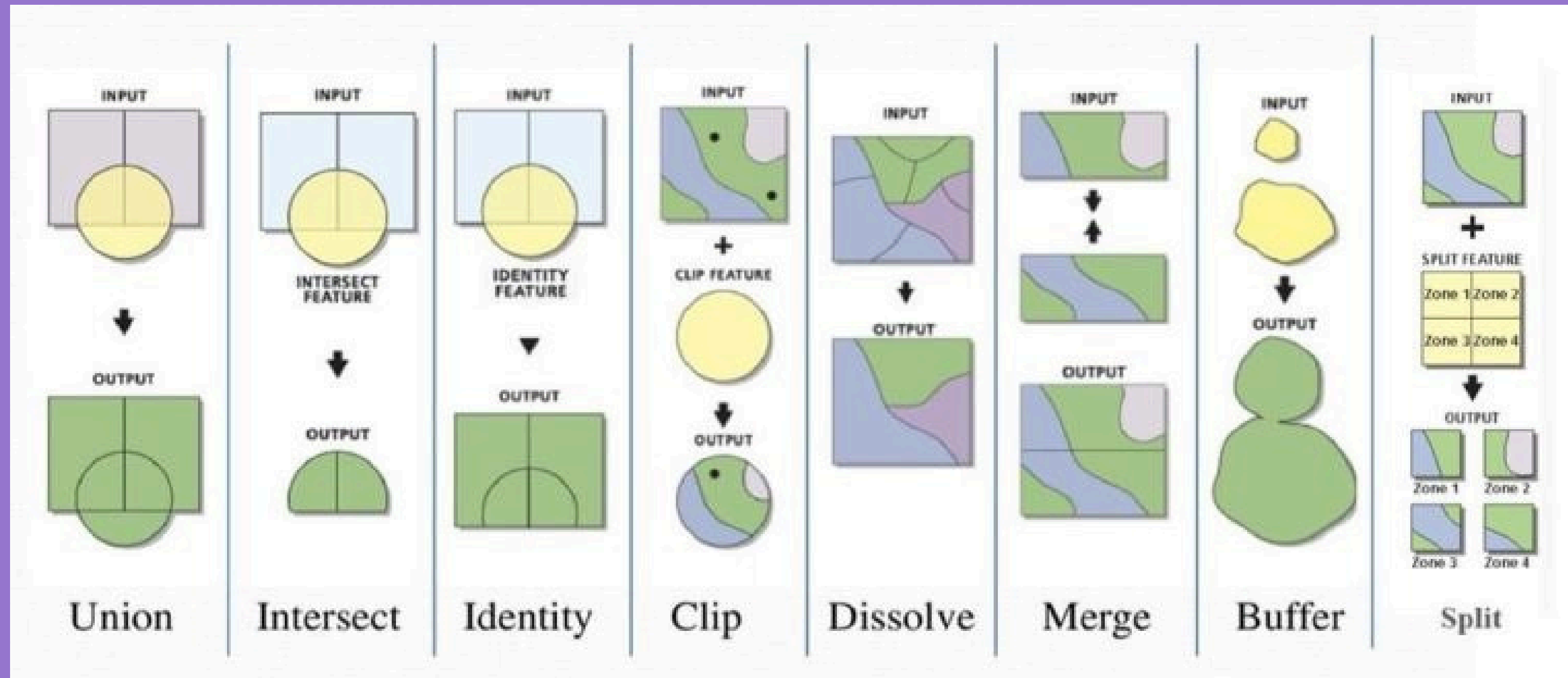
3.Panels

4.Map Canvas

5.Status Bar

# Geoprocessing

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What's the difference?

Code the Map  
Decode the Future

CLOUD  
CLOUD  
CLOUD  
CLOUD

GIS ?

DESKTOP  
DESKTOP  
DESKTOP  
DESKTOP



# Basic Differences

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Decode the Future

## CLOUD GIS

Location Analytics

Simple to Medium Data Processing

Business Focused (Expand, etc)

Real Time Processed

Cloud Based

## DESKTOP GIS

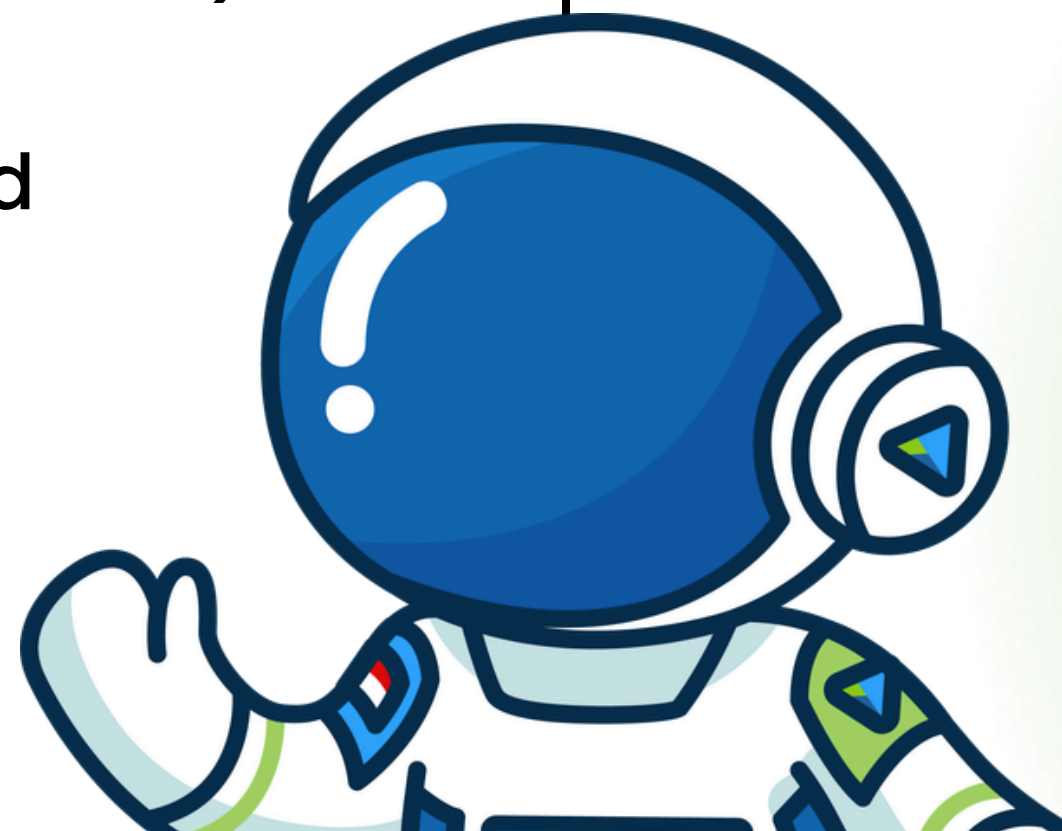
GIS Warehouse Analysis

Simple to Advance Data Processing

Majority on Academic Purposes

Statically Proccesed

Desktop Based





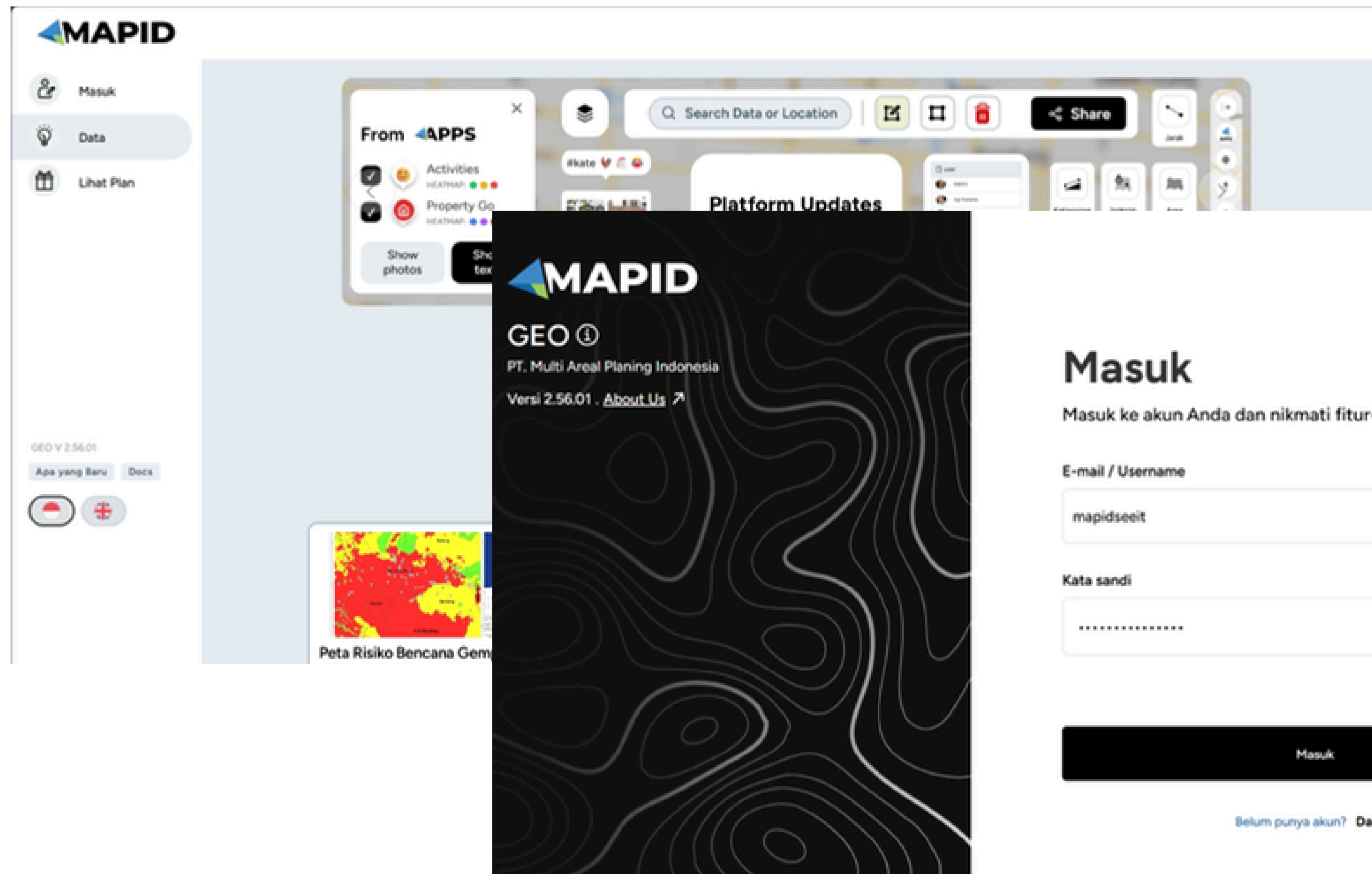
# Introduction to GEO MAPID

Code the Map  
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## GEO MAPID: What can I do?

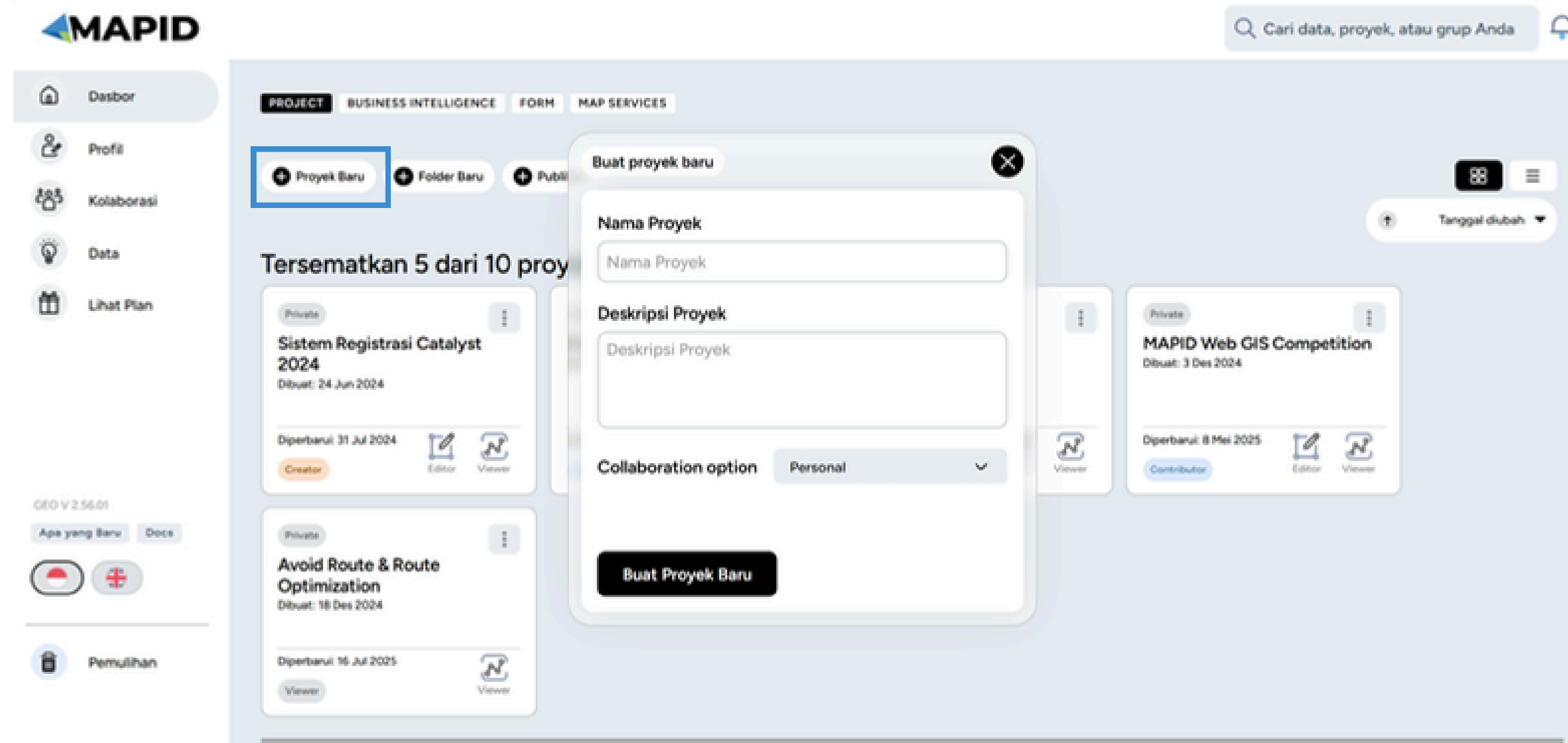
- Creating a custom map by self-digitizing,
- Visualizing existing data,
- Accessing data provided by MAPID,
- Identifying potential locations to expand business,
- Creating a MAPID FORM,
- Making a publication,
- etc.

Code the Map  
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Sign in and generate your idea!

# Pembuatan Project

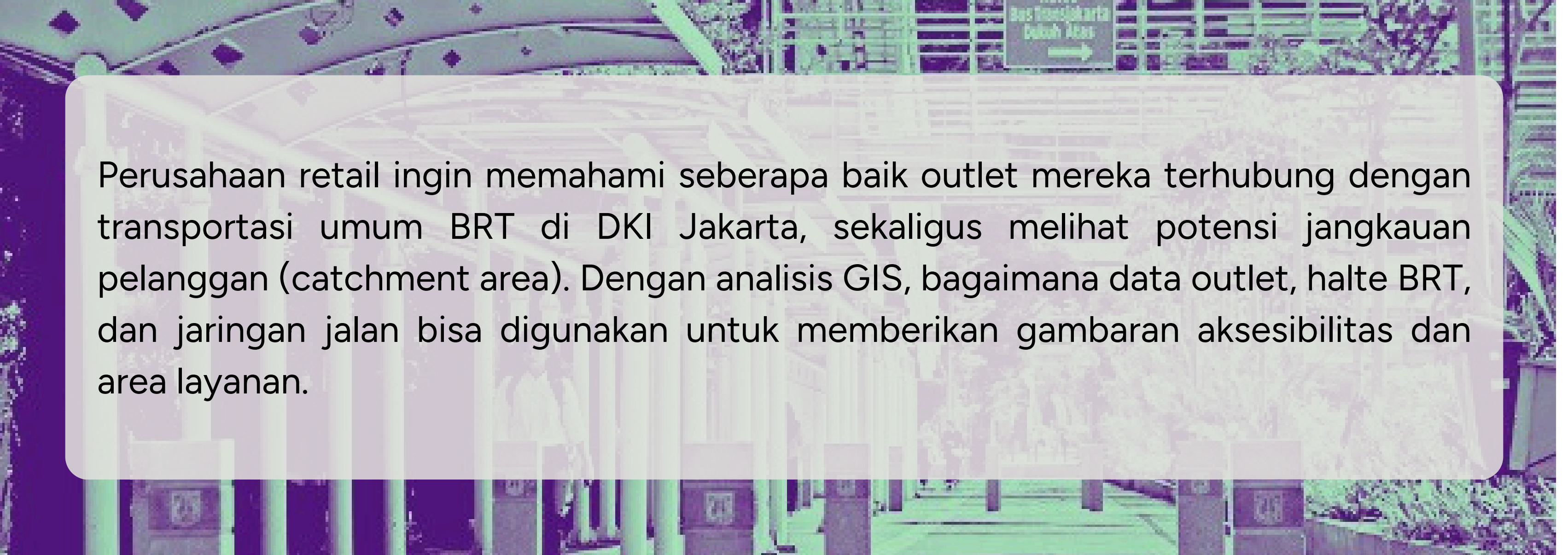


*The journey will begin with one click of this.....*



The screenshot shows the GEO MAPID Map Editor interface. The central map displays Indonesia and surrounding regions. The interface is divided into several sections:

- Left Side Bar:** Contains options for creating and managing layers and data.
  - 1** **Demografi Kota Bandung** (Edit): Includes buttons for "Buat Layer Baru", "Unggah File", "Hubungkan ke API", "Join Table", "Impor data", and "Buat Folder".
  - 2** **Layer mode**: Includes buttons for "Buka folder" and "Tutup folder", a "Nama" field, and a "Filter layer berdasarkan" dropdown.
  - 3** **Multiple Select**: Includes a "Multiple Select" button.
  - 4** **Lainnya**: Includes a "MultiPolygon" button and a layer titled "DEMOGRAFI DI KOTA BANDUNG" by Dikel N J - original.
- Right Side Bar:** Contains options for viewing and managing data.
  - DATA** tab: Includes buttons for "Chat", "History", and "Summary".
  - SINI** tab: Includes a button for "Summary".
  - TOOL** tab: Includes a button for "Summary".
  - 5** **Klik ikon mata untuk melihat data**: Points to the "View" icon in the layer management section.
  - 6** **Right Side Bar:** Includes a list of icons for "Max/min the map", "Your Location", "GEO MAPID Tool", "Legend", "Basemap style", and "Language".



Perusahaan retail ingin memahami seberapa baik outlet mereka terhubung dengan transportasi umum BRT di DKI Jakarta, sekaligus melihat potensi jangkauan pelanggan (catchment area). Dengan analisis GIS, bagaimana data outlet, halte BRT, dan jaringan jalan bisa digunakan untuk memberikan gambaran aksesibilitas dan area layanan.

[mapid.co.id/HandsOn1B3](https://mapid.co.id/HandsOn1B3)



# Home Task

## GIS Fundamental

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[mapid.co.id/Task1B3](https://mapid.co.id/Task1B3)

